

DATABASES: PROTEIN AND SMALL MOLECULES

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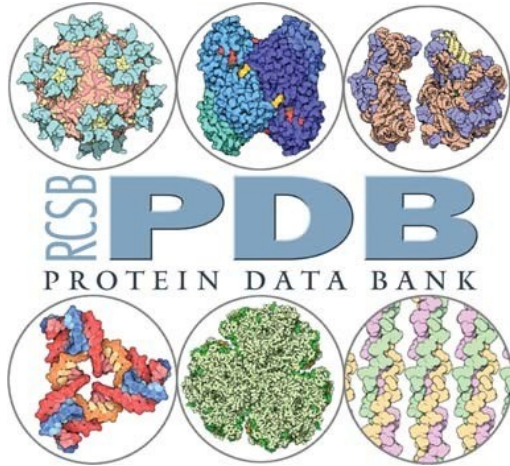
Using Binding DB for drug discovery

Coconut database, Super-Natural 3.0, Natur-AR, Afro-Database

Use and Retrieval of compounds from Drug bank

Chemical space and diversity

The protein Data Bank



~ 250k entries

Search Button

- Repository of all solved structures of proteins, lipids and DNA etc
- Archived in 4 unique code-naming system e.g. **1AKI** for Hen Egg-White Lysozyme

The protein Data Bank

Structure Summary | Structure | Annotations | Experiment | Sequence | Genome | Ligands | Versions

Biological Assembly 1

9NSR | pdb_00009nsr

Crystal structure of bifunctional farnesyl/geranylgeranyl pyrophosphate synthase (FPPS/GGPPS) from Plasmodium falciparum in complex with MMV019313

PDB DOI: <https://doi.org/10.2210/pdb9NSR/pdb>

Classification: **TRANSFERASE**
Organism(s): Plasmodium falciparum 3D7
Expression System: Escherichia coli BL21(DE3)
Mutation(s): No

Deposited: 2025-03-17 Released: 2025-03-26
Deposition Author(s): Seattle Structural Genomics Center for Infectious Disease, Seattle Structural Genomics Center for Infectious Disease (SSGCI)
Funding Organization(s): National Institutes of Health/National Institute Of Allergy and Infectious Diseases (NIH/NIAD), National Institutes of Health/Office of the Director

Experimental Data Snapshot
Method: X-RAY DIFFRACTION
Resolution: 3.25 Å
R-Value Free: 0.295 (Depositor), 0.293 (DCC)
R-Value Work: 0.251 (Depositor), 0.252 (DCC)
R-Value Observed: 0.253 (Depositor)
Starting Model: experimental
View more details

wwPDB Validation
Metric | Percentile Ranks | Value
Rfree | 0.293
Clashscore | 4
Ramachandran outliers | 0
Sidechain outliers | 0.6%
RSRZ outliers | 2.4%
Ligand Structure Quality Assessment

Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (A1B0W)

Global Symmetry: Cyclic - C2 (Explore in 3D)
Global Stoichiometry: Homo 2-mer - A2

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PISA (software)

Literature | Download Primary Citation

Crystal structure of bifunctional farnesyl/geranylgeranyl pyrophosphate synthase (FPPS/GGPPS) from Plasmodium falciparum in complex with MMV019313

[Lovell, S., Cooper, A., Liu, L., Battaile, K.P.](#)

To be published.

Display Files | Download Files | Data API

FASTA Sequence

PDBx/mmCIF Format

PDBx/mmCIF Format (gz)

BinaryCIF Format (gz)

Legacy PDB Format

Legacy PDB Format (gz)

More on PDB File Formats

PDBML/XML Format (gz)

Structure Factors (CIF)

Structure Factors (CIF - gz)

Validation Full (PDF - gz)

Validation (XML - gz)

Validation (CIF - gz)

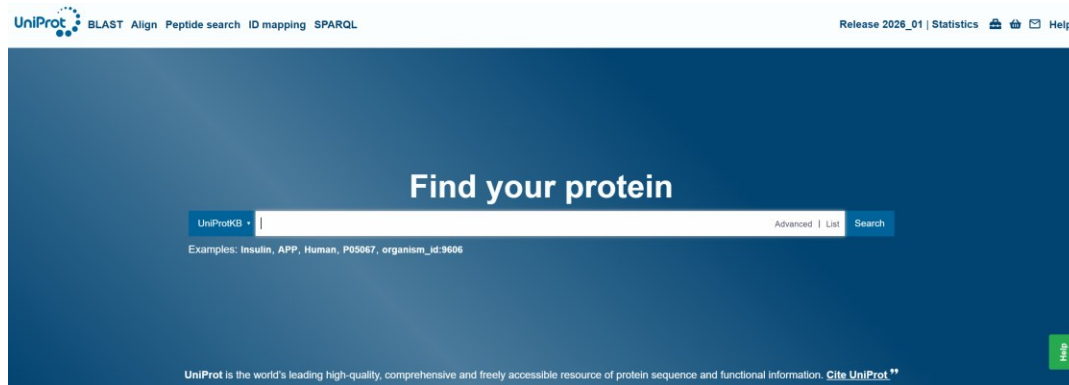
Validation 2fo-fc coefficients (CIF - gz)

Validation fo-fc coefficients (CIF - gz)

Biological Assembly 1 (CIF - gz)

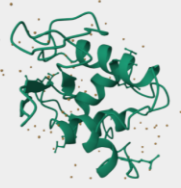
Download options

Uniprot Database



Structureⁱ

View UniProt features on this structure in the [Feature Viewer](#).



SOURCE	IDENTIFIER	METHOD	RESOLUTION	CHAIN	POSITIONS	LINKS
PDB	132L	X-ray	1.80 Å	A	19-147	PDB · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	193L	X-ray	1.33 Å	A	19-147	PDB · RCSB-PDB · PDBj · PDBsum · Foldseek
PDB	194L	X-ray	1.40 Å	A	19-147	PDB · RCSB-PDB · PDBj · PDBsum · Foldseek

P00698 · LYSC_CHICK

Proteinⁱ Lysozyme C
Geneⁱ LYZ
Statusⁱ UniProtKB reviewed (Swiss-Prot)
Organismⁱ Gallus gallus (Chicken)

Amino acids 147 (go to sequence)
Protein existenceⁱ Evidence at protein level
Annotation scoreⁱ

Entry Variant viewer **Feature viewer** **Genomic coordinates** **Publications** **External links** **History**

Tools **Download** **Add** **Community curated (2)** **Add a publication** **Entry feedback**

Functionⁱ
Lysozymes have primarily a bacteriolytic function; those in tissues and body fluids are associated with the monocyte-macrophage system and enhance the activity of immunoagents. Has bacteriolytic activity against M.luteus. **PROSITE-ProRule Annotation** **1 Publication**

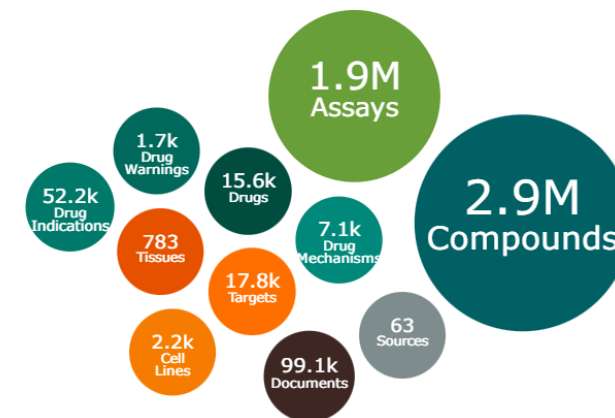
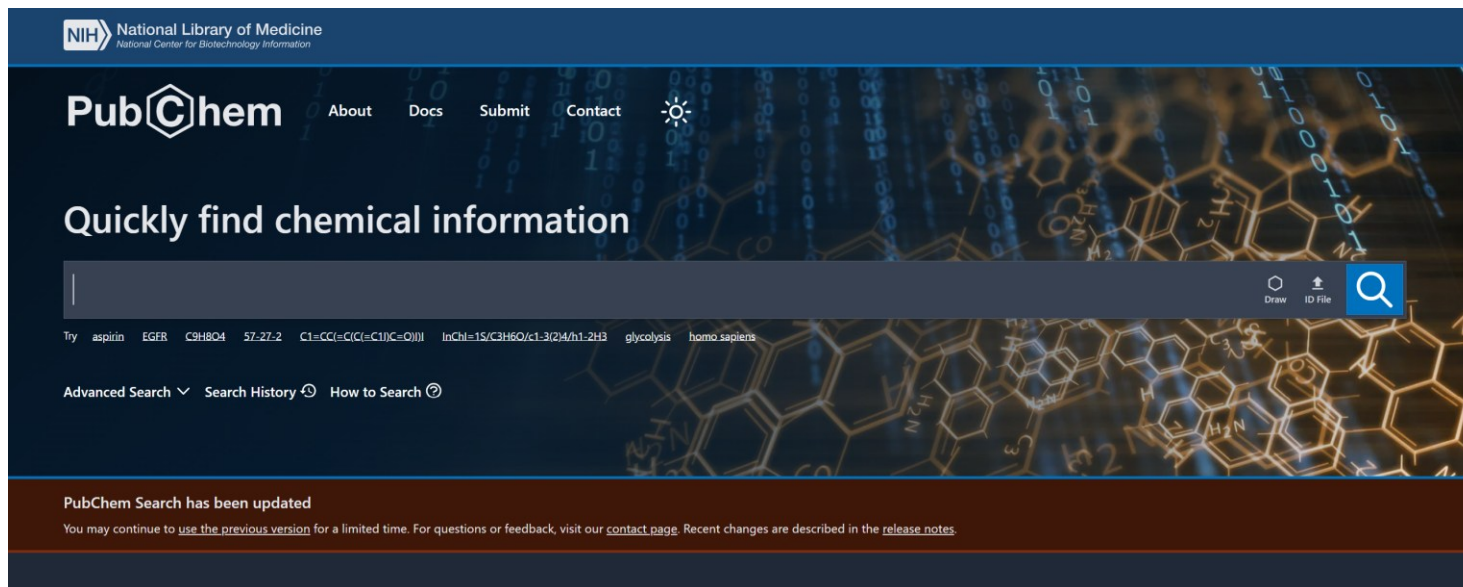
Miscellaneous
Lysozyme C is capable of both hydrolysis and transglycosylation; it also shows a slight esterase activity. It acts rapidly on both peptide-substituted and unsubstituted peptidoglycan, and slowly on chitin oligosaccharides.

Catalytic activityⁱ
Hydrolysis of (1->4)-beta-linkages between N-acetylmuramic acid and N-acetyl-D-glucosamine residues in a peptidoglycan and between N-acetyl-D-glucosamine residues in

- Most comprehensive protein database
- Provides protein sequences with functional annotations from experiments
- A great tool for Bioinformatics

Easily curate list of PDB structures of your protein

PubChem and ChEMBL



- **Most used ligand and bioactivity databases**
- **A great tool for drug discovery and machine learning**

Python API for PubChem and ChEMBL

```
import requests

PUBCHEM_BASE = "https://pubchem.ncbi.nlm.nih.gov/rest/pug"

def get_antiplasmodial_compounds(aid: int = 1159554):
    """Retrieve compounds with antiplasmodial activity from PubChem BioAssay."""
    url = f"{PUBCHEM_BASE}/assay/aid/{aid}/JSON"
    response = requests.get(url)
    response.raise_for_status()
    data = response.json()

    compounds = []
    assay = data["PC_AssayContainer"][0]["assay"]
    for row in assay["results"]["rows"]:
        compounds.append({
            "cid": row["cid"],
            "activity_outcome": row["activity_outcome"]
        })
    return compounds
```

- Retrieve all compounds with **antiplasmodial activity** from **PubChem**
- **PubChem entries** are arranged in assay ID (**AIDs**)

```
import requests

CHEMBL_BASE = "https://www.ebi.ac.uk/chembl/api/data"

def get_antiplasmodial_compounds(limit: int = 100):
    """Retrieve compounds with antiplasmodial activity from ChEMBL."""
    url = f"{CHEMBL_BASE}/activity.json?target_organism=Plasmodium+falciparum&limit={limit}"
    response = requests.get(url)
    response.raise_for_status()
    data = response.json()

    compounds = []
    for act in data.get("activities", []):
        if act.get("standard_type") in ["IC50", "EC50"]:
            compounds.append({
                "molecule_chembl_id": act["molecule_chembl_id"],
                "activity_type": act["standard_type"],
                "value": act["standard_value"],
                "units": act["standard_units"]
            })
    return compounds
```

- Retrieve all compounds with **antiplasmodial activity** from **ChEMBL**
- **Retrieve only compounds with IC50 and EC50 activities**

Using BindingDB

UC San Diego
SKAGGS SCHOOL OF PHARMACY
AND PHARMACEUTICAL SCIENCES

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BindingDB The first public molecular recognition database, BindingDB supports research, education and practice in drug discovery, pharmacology and related fields.

BindingDB contains 3.2M data for 1.4M Compounds and 11.4K Targets. Of those, 1.6M data for 748K Compounds and 4.8K Targets were curated by BindingDB curators. BindingDB is a [FAIRsharing](#) resource.

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BindingDB Data by Target

Target Names are taken from UniProtKB. To view the associated UniProt page, click on the UniProt ID. When we have data for a given Target in multiple species or for mutants of a given Target, we provide a header row (upper case, bold font) with all of the data followed by a separate for each species or variant of the Target.

Target Name contains [Search](#)

All 0-9 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

Target Name	Organism	Affinity Links	Articles	UniProts	Files
A disintegrin and metalloproteinase with thrombospondin motifs 1 Q9UHI8	Human	Ki: 5 IC50: 22	4	1	2D 3D TSV
A disintegrin and metalloproteinase with thrombospondin motifs 13 Q76LX8	Human	IC50: 4	3	1	2D 3D TSV
A DISINTEGRIN AND METALLOPROTEINASE WITH THROMBOSPONDIN MOTIFS 4 A0AB13Q3A2 O75173		Ki: 13 IC50: 610 Other: 2	38	2	
A disintegrin and metalloproteinase with thrombospondin motifs 4 O75173	Human	Ki: 13 IC50: 604 Other: 2	38	1	2D 3D TSV
ADAM metallopeptidase with thrombospondin type 1 motif 4 A0AB13Q3A2	Dog	IC50: 6	1	1	2D 3D TSV

Targets ▼
Compounds ▼
Publications ▼
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Special Tools ▼
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Tutorials
myBDB

**Search across
targets and
compounds**

A great resource for well-curated activity dataset for machine learning, QSAR modelling and benchmarking of algorithms

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Organism

Citations

Q Search compound name, SMILES, InChI, InChI Key

Q Search

Try: [Caffeine](#), [CNP0228556](#)

Report bugs: [Issue Tracker](#)



Draw Structure

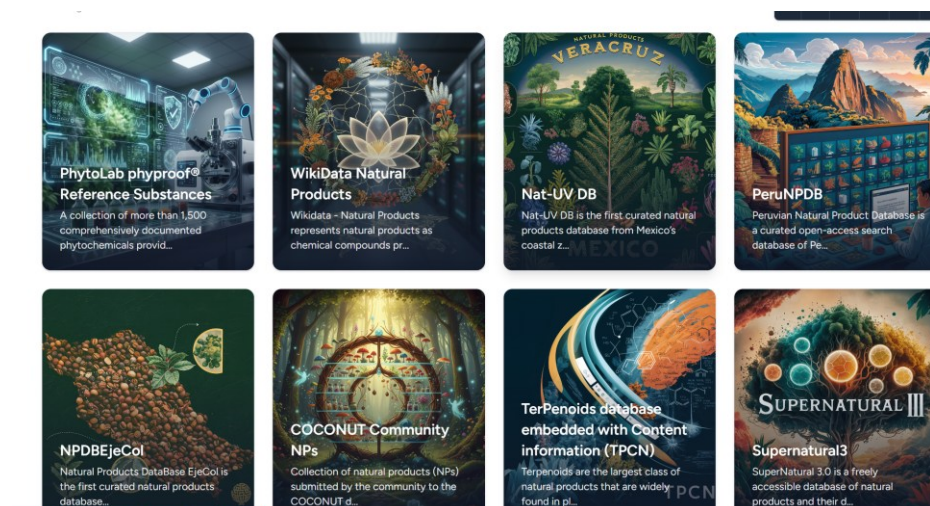


Browse Data



Submit Data

- Greatest collection of natural products in different activity classes
- Up to a million compounds available
- Easy to retrieve in csv format with smiles notations or sdf format



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ASK AI

What drugs are in development that target proteins containing an SH2 domain?

Next Question

You can request for sdf file of all approved and experimental drugs via the academic license for your drug discovery purposes

Approved drug

Zidovudine

Remdesivir

Carbamazepine

Drug repurposing

In silico-based approach

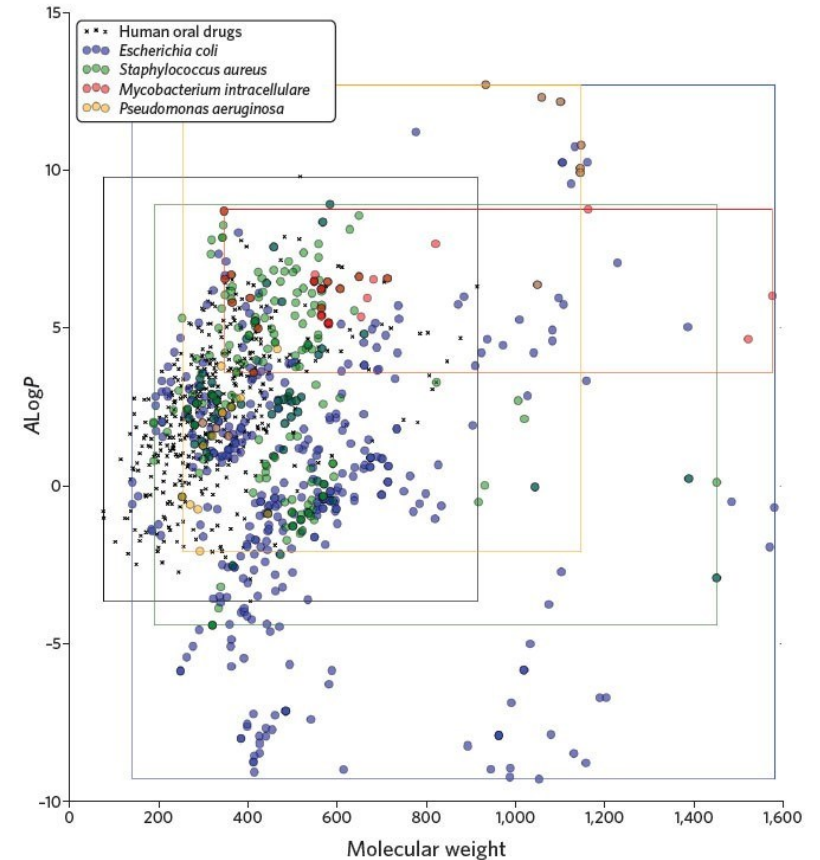
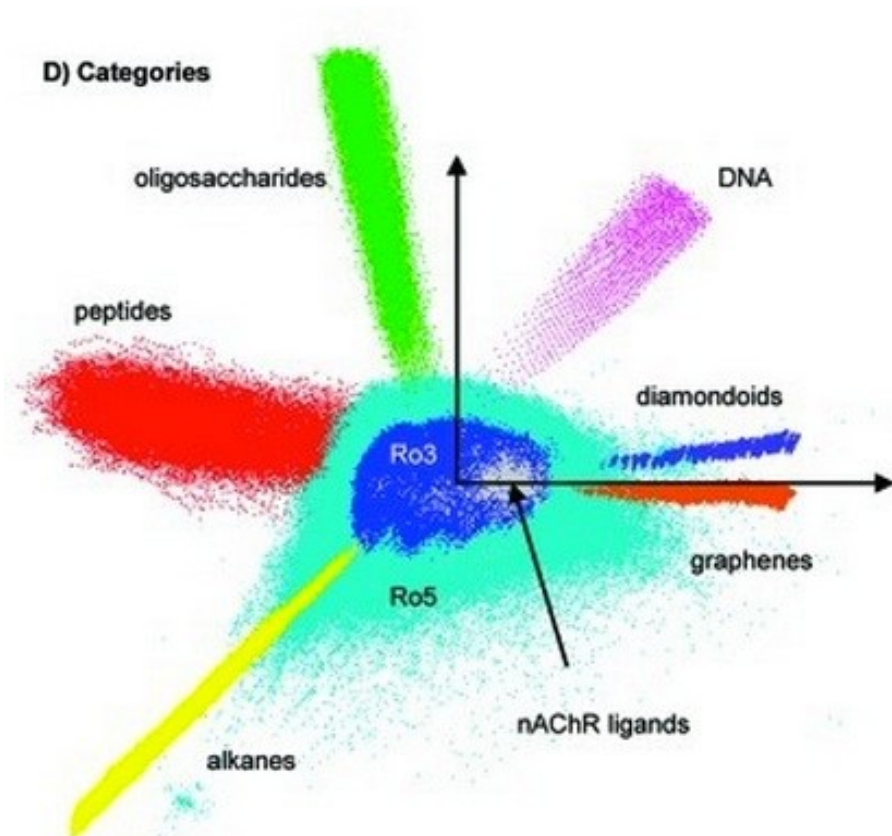
Repurposed indication

HIV/AIDS

COVID-19

Trigeminal neuralgia

The Chemical Space



- A way to answer “**How many compounds do we have?**”
- **A way to characterize the compounds in our library**
- Where is each compound in the property-space?

LogP – MW plot is the easiest to visualize

Chemical Libraries

Major Library players

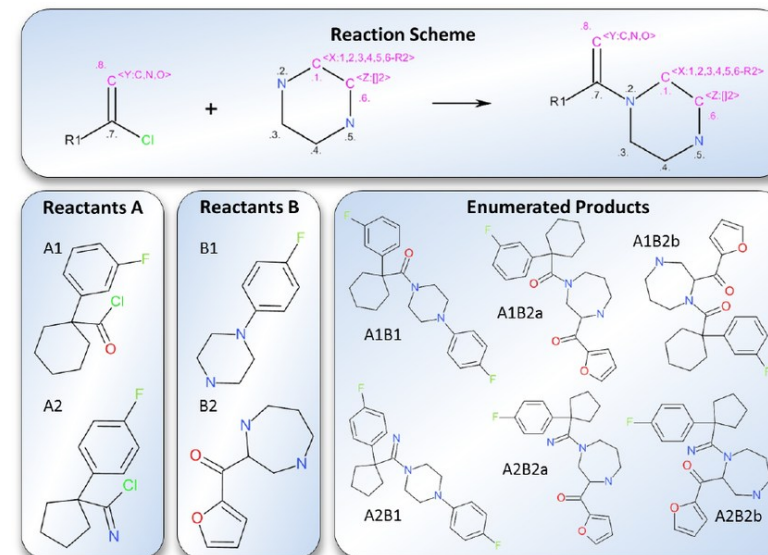
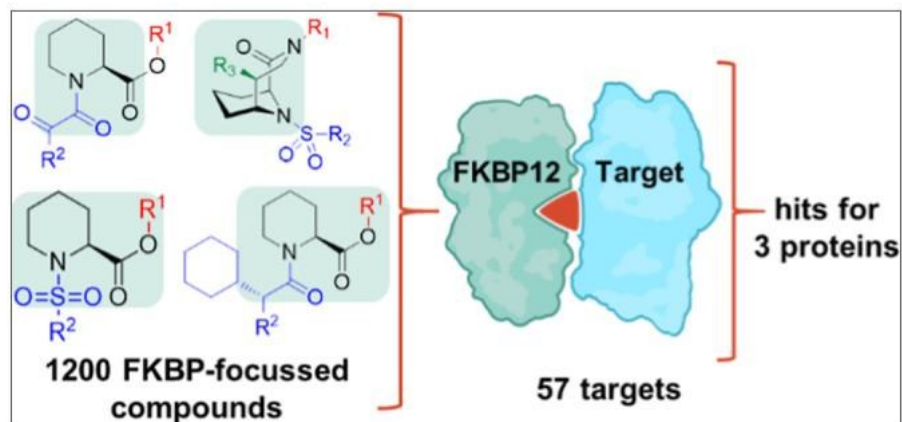


eMolecules

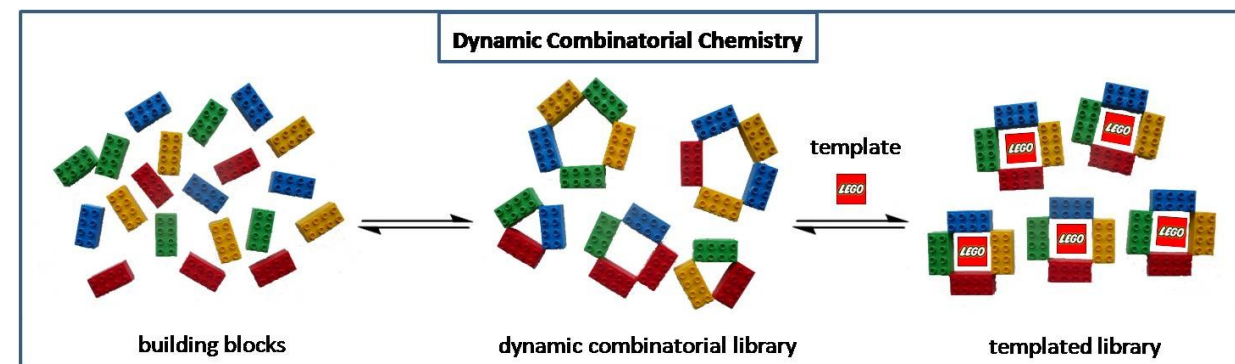
PharmaBlock



Focused Library



Enumerated Library



Further References

- <https://www.rcsb.org/>
- <https://www.uniprot.org/>
- <https://www.bindingdb.org/rwd/bind/index.jsp>
- <https://go.drugbank.com/>
- <https://coconut.naturalproducts.net/>
- <https://www.ebi.ac.uk/chembl/>